Literal equations and formulas

1 Solve for x in the following equations:

a $y = x + z$

b $y = \frac{x}{4}$

c $y = -8\,(7 + x)$

d $\frac{x}{9} + \frac{n}{2} = 5$

e $-\frac{x}{9} - \frac{n}{5} = 5$

f $kx = m + nx$

2 Solve for m in the following equations:

a $\frac{m}{y} = gh$

b $y = 9mx - 5$

c $x = -2k\,(n + m)$

d $x = 3k\,(n + m)$

3 Solve for k in the following equations:

a $r = \frac{k}{k - 3}$

b $m = \frac{5kx}{7} - 3$

4 Solve for y : $\frac{x}{y} = h - k$

5 Solve for R in the following equations:

a $V = IR - E$

b $D = RT$

6 The formula that converts Celsius into Fahrenheit is $F = \frac{9}{5}C + 32$.
Explain how would you rearrange this formula to solve for C.

7 The bend allowance for sheet metal is given by the formula $B = 2\pi\left(R + \frac{T}{2}\right)\frac{A}{360}$.
Solve for T, the thickness of the sheet.

8 The displacement of an object is given by $S = \frac{1}{2}at^2 + v_0 t$.
Solve for the initial velocity v_0.

9 The formula $a = \frac{v_f - v_i}{t}$ is used to calculate acceleration a as the change in velocity over the period of time t.
Solve the formula for final velocity, v_f.

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10 The formula to convert from Fahrenheit to Celsius is defined by $C = \frac{5}{9}(F - 32)$.
The temperature inside a freezer is 86°F. Find this temperature in Celsius.

11 Sally bought a television series online. The television series was 7.2 gigabytes large and took 2 hours to download.

a Determine the size of the television series in gigabits. Use the formula $b = 8B$, where b is the number of gigabits and B is the number of gigabytes.

b Hence, calculate the average number of gigabits downloaded each hour.

12 The area of triangle, A, is defined by formula $A = \frac{b \times h}{2}$ where b is the base and h is the height.
Find the height of a triangle with Base $= 5$ and Area $= 30$.

13 The volume of a prism is given by the formula $V = a \times b \times h$.
Find h if $a = 2$, $b = 6$ and $V = 108$.

14 The surface area of the rectangular prism is given by formula $S = 2(ab + bh + ah)$.
Find the value of h when $a = 4$, $b = 8$ and $S = 304$.

15 The power of a circuit is given by the formula $P = I^2R$.
Find the value of I when $P = 80$ and $R = 1280$. Round your answer to two decimal places.

16 Newton's second equation of motion is $s = ut + \frac{1}{2}at^2$.
Find the value of a if $s = 4000$, $u = 300$ and $t = 20$.

17 The resistance of a parallel circuit is given by the formula $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$.
Find the value of b if $a = 12$ and $c = 27$. Round your answer to two decimal places.

18 Write down an equation for the following statement:
'The revenue (R) of a company is double the cost (C).'

19 The volume of a rectangular prism is given by the formula $V = l \times w \times h$, where l , w and h are the dimensions of the prism.
Given that a rectangular prism has a length of 4 cm, a width of 8 cm and a height of 5 cm, find its volume.

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20 Given that the surface area of a cylinder $= 2\pi r^2 + 2\pi rh$, find the exact surface area of a cylinder with diameter $10\ \text{cm}$ and height $23\ \text{cm}$.

21 The life expectancy, y, of a person born in Japan is approximated by the equation $y = 0.28x + 73$, where x is the number of years since 1970.

- **a** Solve for x.
- **b** Find the value of x where life expectancy will be 95, correct to three decimal places.
- **c** In which year will the life expectancy in Japan reach 95 years?

22 The amount of oil, y, imported to the United States from Canada, in millions of barrels per day, can be approximated by the equation $y = 0.106x + 1.37$, where x is the number of years since 2000.

- **a** Solve this equation for x.
- **b** Hence, determine x, the number of years after 2000 that the approximate number of oil barrels imported from Canada per day will be 3.5 million. Round your answer to three decimal places.
- **c** Hence, in which year will the approximate number of barrels imported from Canada reach 3.5 million per day?

Additional questions

23 What benefits come from rewriting literal equations?

24 The formula for the force of an object is $F = ma$, where F is the force, m is the mass of the object, and a is the object's acceleration. Write an equation that could be used to solve for the mass of the object.

25 The formula for the perimeter of a rectangle is $P = 2l + 2w$, where P is the perimeter, l is the length of the rectangle, and w is the width. Write an equation that could be used to find the length of the rectangle.

26 Betsy knows the formula to convert Celsius into Fahrenheit is $F = \frac{9}{5}C + 32$. She wants to convert a temperature from Fahrenheit to Celcius. Find an equation that she could use to do this conversion.

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a The formula to the distance traveled in a time period is $d = rt$, where d is the distance traveled, r is the rate of travel, and t is the time spent traveling. Write an equation that could be used to solve for the rate of travel.

b Floyd is riding his bike 5 miles to Jue's house. Jue is expecting Floyd in 15 minutes. What is the average speed Floyd must ride in order to get to Jue's on time?

28 Determine the similarities and differences of solving $3x + y = z$ and $3x + 8 = 2$. Explain your thinking.

29 Rufino has submitted the following work to write the equation for the mass of an object, m, given the kinetic energy, KE, and velocity, v:

$$\begin{aligned} KE &= \frac{1}{2}mv^2 \\ 2KE &= mv^2 \\ \sqrt{2KE} &= mv \\ \frac{\sqrt{2KE}}{v} &= m \\ m &= \frac{\sqrt{2KE}}{v} \end{aligned}$$

Determine whether Rufino is correct by explaining each step of his working if the step is correct, otherwise fix his error.

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